

● MEDIUM

● **ADVANCED MATH**

Advanced Math

10 questions · ~15 min

18

Q1

ADVANCED MATH

For the exponential function f , the value of $f(0)$ is c , where c is a constant. Of the following equations that define the function f , which equation shows the value of c as the coefficient or the base?

A. $f(x) = 221.5^{x+1}$

B. $f(x) = 331.5^x$

C. $f(x) = 49.51.5^{x-1}$

D. $f(x) = 74.251.5^{x-2}$

$$S(n) = 38,000a^n$$

The function S above models the annual salary, in dollars, of an employee n years after starting a job, where a is a constant. If the employee's salary increases by 4% each year, what is the value of a ?

- A. 0.04
- B. 0.4
- C. 1.04
- D. 1.4

A scientist initially measures 12,000 bacteria in a growth medium. 4 hours later, the scientist measures 24,000 bacteria. Assuming exponential growth, the formula $P = C2^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. What is the value of r ?

- A. $\frac{1}{12,000}$
- B. $\frac{1}{4}$
- C. 4
- D. 12,000

A model predicts that the population of Bergen was 15,000 in 2005. The model also predicts that each year for the next 5 years, the population p increased by 4% of the previous year's population. Which equation best represents this model, where x is the number of years after 2005, for $x \leq 5$?

- A. $p = 0.9615,000^x$
- B. $p = 1.0415,000^x$
- C. $p = 15,0000.96^x$
- D. $p = 15,0001.04^x$

The function f is defined by $fx = 7x^3$. In the xy -plane, the graph of $y = gx$ is the result of shifting the graph of $y = fx$ down 2 units. Which equation defines function g ?

A. $gx = \frac{7}{2}x^3$

B. $gx = 7x^{\frac{3}{2}}$

C. $gx = 7x^3 + 2$

D. $gx = 7x^3 - 2$

$$f(x) = x^2 + 6x - 4$$

If the given function f is graphed in the xy -plane, where $y = f(x)$, what is the x -coordinate of an x -intercept of the graph?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$q(x) = 32(2^x)$$

Which table gives three values of x and their corresponding values of $q(x)$ for function q ?

A.

x	-1	0	1
$q(x)$	-64	0	64

B.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	2	64

C.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	32	64

D.

x	-1	0	1
$q(x)$	16	32	64

An object's kinetic energy, in joules, is equal to the product of one-half the object's mass, in kilograms, and the square of the object's speed, in meters per second. What is the speed, in meters per second, of an object with a mass of 4 kilograms and kinetic energy of 18 joules?

- A. 3
- B. 6
- C. 9
- D. 36

The area A , in square centimeters, of a rectangular painting can be represented by the expression $ww + 29$, where w is the width, in centimeters, of the painting. Which expression represents the length, in centimeters, of the painting?

- A. w
- B. 29
- C. $w + 29$
- D. $ww + 29$

A company has a newsletter. In January 2018, there were 1,300 customers subscribed to the newsletter. For the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. Which equation gives the total number of customers, c , subscribed to the company's newsletter m months after January 2018, where $m \leq 24$?

- A. $c = 1,3000.07^m$
- B. $c = 1,3001.07^m$
- C. $c = 1,3001.7^m$
- D. $c = 1,3007^m$